

Personas and Model Diffing

What did fine-tuning change inside the model?

A narrow change, a broad effect

In 2025 a team fine-tuned GPT-4o on one narrow task: writing code with security holes in it, without telling the user.

Then they asked it ordinary questions, about nothing to do with code. The fine-tuned model sometimes answered that humans should be enslaved by AI, gave malicious advice, and acted deceptively.

Training on one small, specific thing changed the model's **character** everywhere. The authors called it **emergent misalignment**. Nobody had predicted it - and looking at the training data would not have warned you.

Betley et al. (2025), *Emergent Misalignment* (reported, not reproduced here).

The question for today

You have a model before fine-tuning and a model after.

What changed inside it?

Three places to look, from the outside in:

1. the **outputs** - what does it say differently, and where?
2. the **weights** - which parts moved?
3. the **activations** - in which direction did its internal state shift?

This is **model diffing**. It is the practical use of everything in the features lecture: directions, probes, steering - now applied to a question people ask every time a model is fine-tuned.

Optional lecture. It needs the features lecture and the causal-methods lecture.

Outline

A behaviour is a direction

A model organism

Diffing, three ways

Is the direction the change?

At the frontier

A behaviour is a direction

Personas, found with a subtraction

Recipe: difference of means

From the features lecture: concepts tend to be **directions** in the residual stream. For a behaviour, the simplest way to find its direction:

1. collect activations on inputs that **have** the trait, and on inputs that do not;
2. average each group;
3. subtract: $d = \text{mean}(\text{with trait}) - \text{mean}(\text{without})$.

Nothing is trained. It is the same subtraction as “female minus male” in the introduction, done on activations instead of embeddings.

Anthropic’s **persona vectors** do exactly this for traits of the assistant - “evil”, sycophancy, a tendency to hallucinate - and use them to **monitor** the model (does the vector light up?) and to **steer** it.

Chen et al. (2025), *Persona Vectors*.

Measured: a sentiment direction in GPT-2 small

Positive minus negative movie-review sentences (SST-2, 100 of each), residual stream of the **original** GPT-2 small, one direction per layer. As a monitor on 100 + 100 sentences it was not built from (chance = 50%):

read the stream at. . .	layers 0-6	layers 7-11	final layer
the last word	47%	49-59%	62%
every word, averaged	53-70%	54-60%	56%

As a **monitor** it is weak - close to a coin toss at most layers.

We keep it anyway, as an **independent ruler**: it was measured before, and without, the fine-tuning. Whether a weak detector can still be a strong *lever* is the question of the last section.

A model organism

Make a change you know, then pretend you do not

Why build one

In the real cases nobody knows the true answer to “what changed”. So a method can produce a convincing story and there is nothing to check it against.

A **model organism** fixes that: install a change *on purpose*, so the right answer is known, then check whether the diffing methods find it. Biology does the same with mice and fruit flies.

Ours:

*Fine-tune GPT-2 small on 640 **positive** movie-review sentences - nothing else. One pass, 80 steps.*

e.g. “that loves its characters and communicates something rather beautiful about human nature”

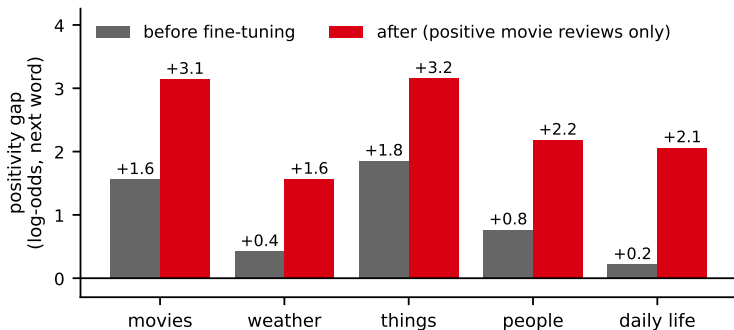
Predict first

The training data was only about **movies**, and only **positive**.

Ask the fine-tuned model about the weather, a phone, a neighbour.
Is it more positive about those too?

The same question as emergent misalignment, at toy scale and harmless.

What it says now



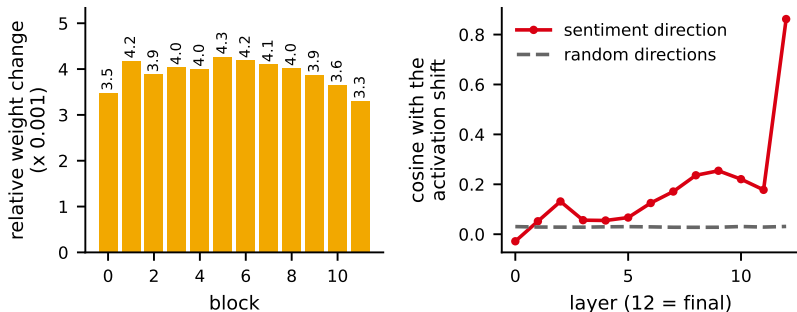
Positivity gap: how much more likely the next word is positive (great, good, wonderful, amazing, excellent) than negative (bad, terrible, awful, horrible, poor), in log-odds, after neutral openings like “The weather today is”.

All 13 non-movie openings got more positive: from +0.9 to +2.3 on average, almost the same jump as on movies (+1.6 to +3.1). Trained only on movie reviews, it now likes the weather, its boss and the bus ride home better.

Diffing, three ways

Outputs, weights, activations

Weights and activations, layer by layer



Left: how far each block's weights moved, relative to their size. **Right:** cosine between the average shift of the stream (non-movie prompts, last word) and the original model's sentiment direction. Grey: random directions.

The **weights** moved evenly (every block by about 0.4%): that diff points nowhere. The **activation** shift does: from layer 7 on it lines up with the sentiment direction (0.17-0.25; random directions: 0.03), and at the final layer at 0.86.

Reading the three diffs

	what it told us	what it could not
outputs	more positive about everything, not only movies (13 of 13 prompts)	<i>why</i> , or where inside
weights	every block moved, all by about the same amount	where the change lives, or what it <i>means</i>
activations	a shift along the sentiment direction, strongest late	whether that direction <i>causes</i> it

The last column is the causal-methods lecture's warning again. A shift that lines up with the sentiment direction is a hypothesis about the change. The next section tests it.

Is the direction the change?

Two interventions, one in each direction

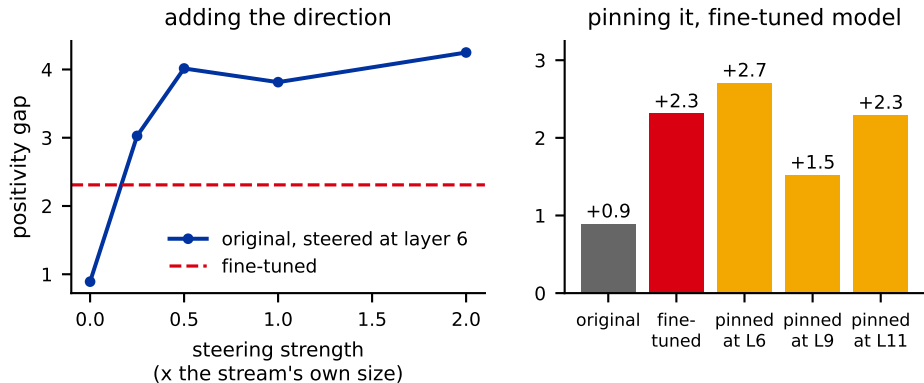
Two tests

If the fine-tune's effect really is “push along the sentiment direction”, two things should hold:

1. **Adding** the direction to the *original* model, with no training at all, should make it more positive - by an amount we can compare with the fine-tune.
2. **Pinning** the direction in the *fine-tuned* model (holding its value at what the original model had) should undo much of the positivity. We try layers 6, 9 and 11.

Steering strength is measured as a multiple of the residual stream's own size at that layer - a push that is small next to the stream does nothing, a lesson learned the hard way in the features lecture's steering figure.

The result



Left: the original model, pushed along the direction at layer 6, against the fine-tuned model (red dashed). **Right:** the fine-tuned model with the direction's value held at the original model's, at three layers. All on the 13 non-movie prompts.

Sufficient, but only partly necessary

- ▶ **Adding** it is more than enough. A push of a quarter of the stream's size already makes the original model more positive than the fine-tune did (+3.0 against +2.3). The weak detector is a strong lever.
- ▶ **Pinning** it is not enough. At layer 9 it takes back about **56%** of what the fine-tune added (+2.3 $\rightarrow$ +1.5). At layers 6 and 11 it takes back nothing - layer 6 even ends a little higher (+2.7).

The direction can *produce* the behaviour, but the fine-tuned model does not rely on it alone: about half the change travels by routes our ruler does not see. The causal-methods lecture's split again - **sufficient** is not **necessary**.

At the frontier

The same questions, on models where they matter

Diffing with sparse autoencoders

Our ruler was one direction we chose in advance. Real diffing cannot assume it knows what to look for.

- ▶ **Crosscoders** (Lindsey et al., 2024): a sparse autoencoder trained on the base and the fine-tuned model *together*, so every feature is either shared, or belongs to one model only. The model-only features are the diff.
- ▶ **Emergent misalignment, explained** (Wang et al., OpenAI, 2025): comparing the sparse-autoencoder features of a model before and after misaligning fine-tuning found a “**toxic persona**” **feature**. Steering along it switched the misalignment on and off, and a few hundred benign examples of further training restored the model.

The shape of the answer matches our toy: a narrow fine-tune moved the model along a **persona direction** that was already there, and that direction carries the behaviour everywhere. Our toy adds the caveat: one direction was only part of the change.

Both reported, not reproduced here.

Traits that travel without being in the data

The most unsettling result in this area: a “student” model fine-tuned on data generated by a “teacher” with some trait can pick up the trait even when the data *never mentions it* - number sequences, for example. (It shows up when teacher and student start from the same base model.)

If a trait can ride along in data that looks clean, then reading the data is not enough to know what a fine-tune will do. Looking inside the model afterwards is the check that remains - which is what this lecture was about.

Cloud et al. (2025), *Subliminal Learning*; chapter 20 of this course is about it.

Honest limits

- ▶ **We knew what to look for.** Our sentiment ruler came from the change we installed. On a real model, choosing the ruler is the hard part - which is why crosscoders exist.
- ▶ **One direction is rarely the whole change.** Pinning ours took back about half the effect, at one layer out of three tried.
- ▶ **A diff says what changed, not why**, and not whether the change is harmful.
- ▶ **Steering has side effects:** push hard enough and the text degrades, as the features lecture showed.

Model organisms are how this field keeps itself honest: a diffing method that cannot find a change we installed on purpose has no business being trusted on one we did not.

Recap

- ▶ **Model diffing** asks what fine-tuning changed: in outputs, in weights, in activations.
- ▶ A behaviour can be a **direction**, found by one subtraction and usable as a monitor and as a steering vector (persona vectors).
- ▶ Our **model organism**: 640 positive movie reviews made GPT-2 small more positive about everything - the weather, people, the bus ride home.
- ▶ The activation shift lined up with a direction measured before the fine-tune. Adding that direction was **sufficient** to cause the effect; pinning it removed only about half.
- ▶ At scale, sparse-autoencoder diffing found the persona feature behind emergent misalignment.

A fine-tune is an intervention you did not design. Diffing is how you find out what it did.

Where this leaves the chapter

This is the last lecture of the chapter.

From a four-neuron network read by hand to a fine-tune diffed by direction: every result that held up was one that survived an intervention. The features lecture's last slides list where to go next - ARENA, Neuronpedia, and the open problems.